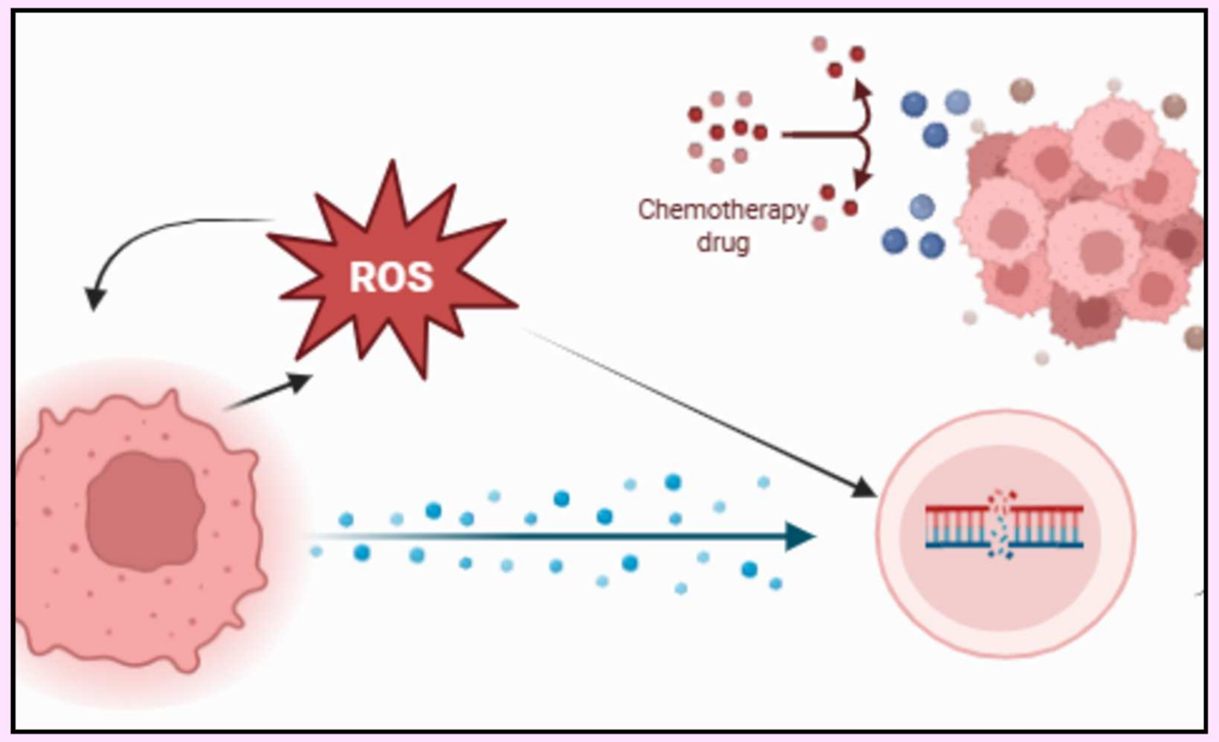
Comparing the Role of Vitamin B7 and Vitamin B12: Impact on IL-6 Expression by 4T1 Breast Cancer Cells Inflamed by Lipopolysaccharides

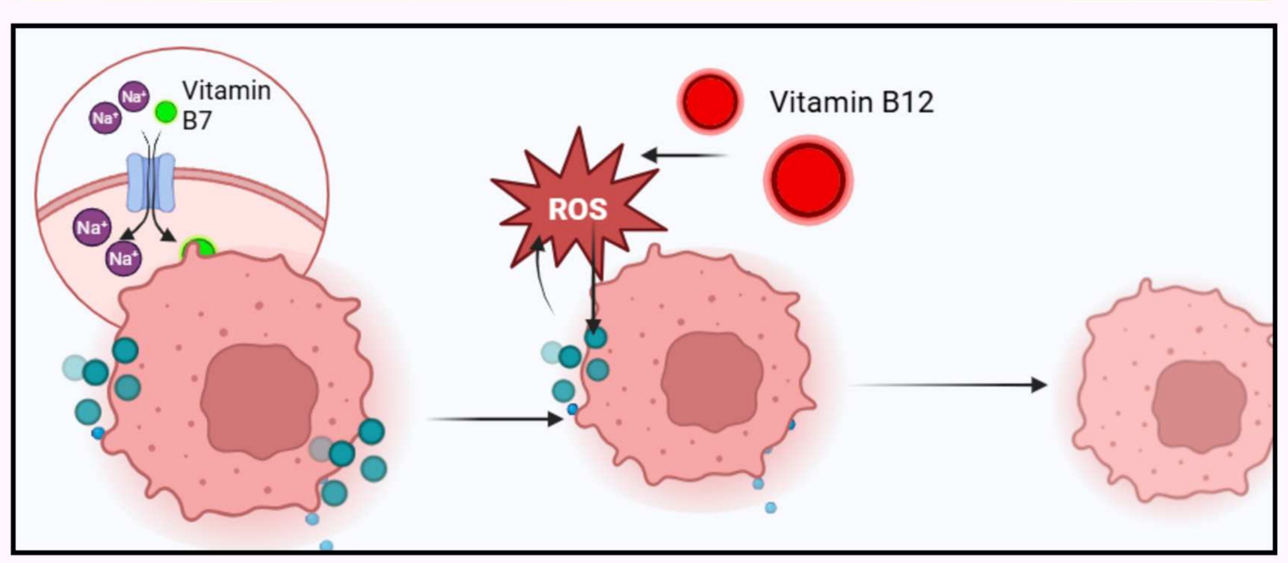
Background



- Breast cancer is the second deadliest cancer, affecting 1 in 8 women in the U.S.. Currently, only 32% of women with the cancer spreading to distant parts of the body survive. (Yale, 2024).
- Current treatments include hormonal therapy, immunotherapy, surgery, radiotherapy, personalized therapy, and chemotherapy.
- Chemotherapy damages healthy cells due to IL-6 resistance (Greten et al.,)

Related Literature

- Vitamin B7 is a drug conjugate used to deliver medication directly to tumor cells, allowing reduced toxicity to healthy cells (Wang et al.,)
- Vitamin B12: 5 antioxidant properties, direct scavenging of ROS (van de Lagemaat et al.,)



Hypothesis

If 4T1 cells are treated with a dual treatment of Vitamin B7 and Vitamin B12, then the expression of IL-6 cytokine proteins will decrease, because Vitamin B7 will target the overexpressed SMVTs, allowing Vitamin B12's antioxidant properties to directly scavenge released reactive oxygen species (ROS).

Methodology

1. Thaw & Culture

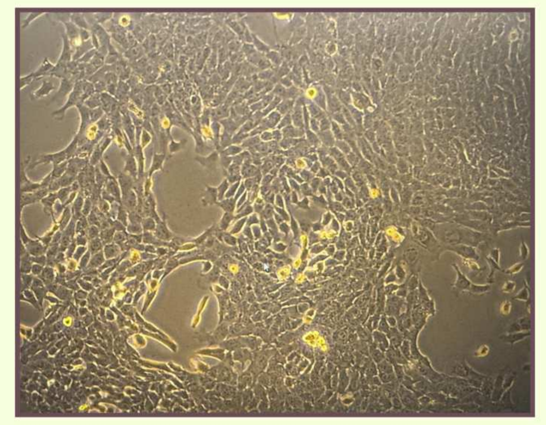
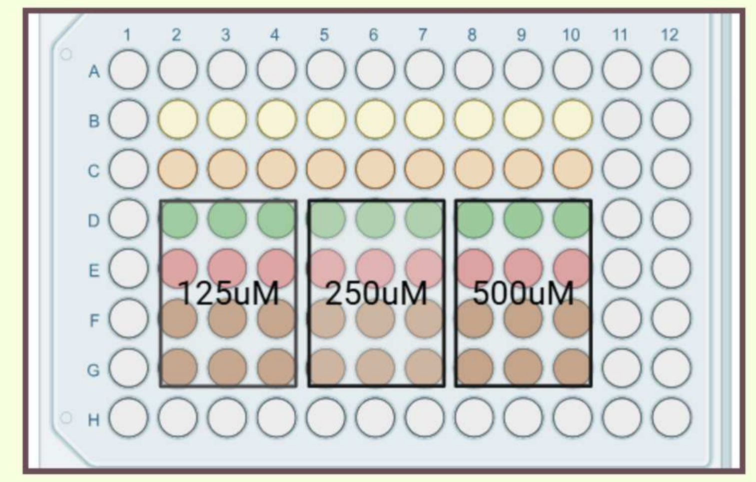


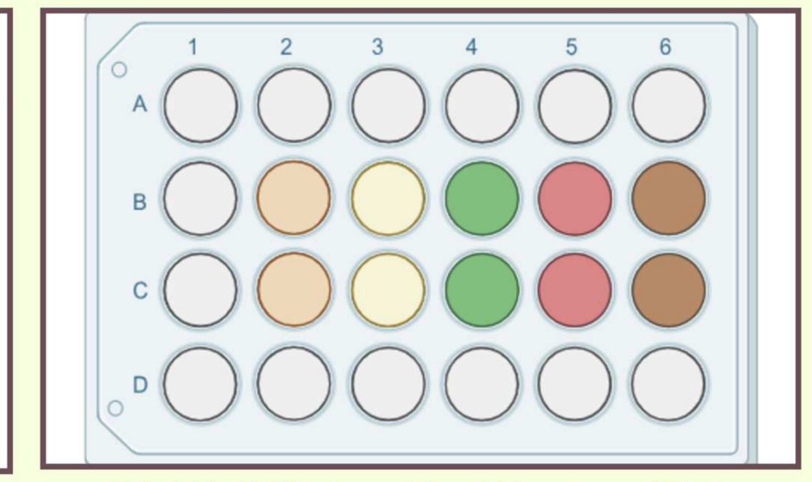
Figure 1. 4T1 Breast Cancer Cells - 1.22.26. - Confluency : 90% - Magnification : 400X

2. Seed and 3. Treat Well Plates

Negative Control | Vitamin B7 Treatment | Dual Treatment
Positive Control (LPS) | Vitamin B12 Treatment | Empty



96-Well Plate = Acid Phosphatase Assay

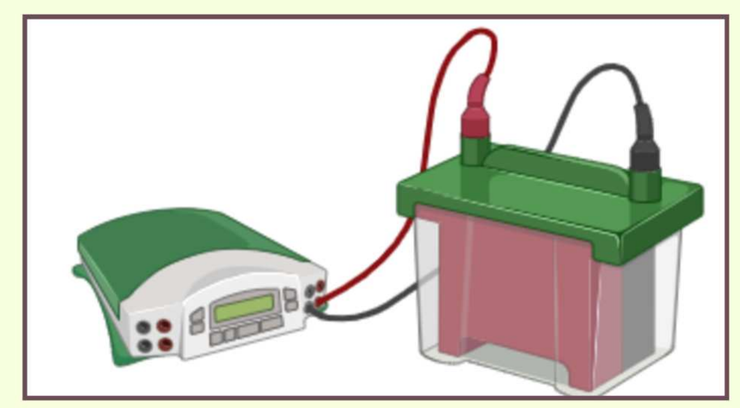


24-Well Plate = For Western Blot Assay

4. Acid Phosphatase

Test for: Cell Viability before Testing Protein Presence

5. Western Blot



Test For: Presence of IL-6 Protein using Chemiluminescence

6. Statisitcal Analysis

Kruskal-Wallis Test with post-Hoc Dunn's test

Results

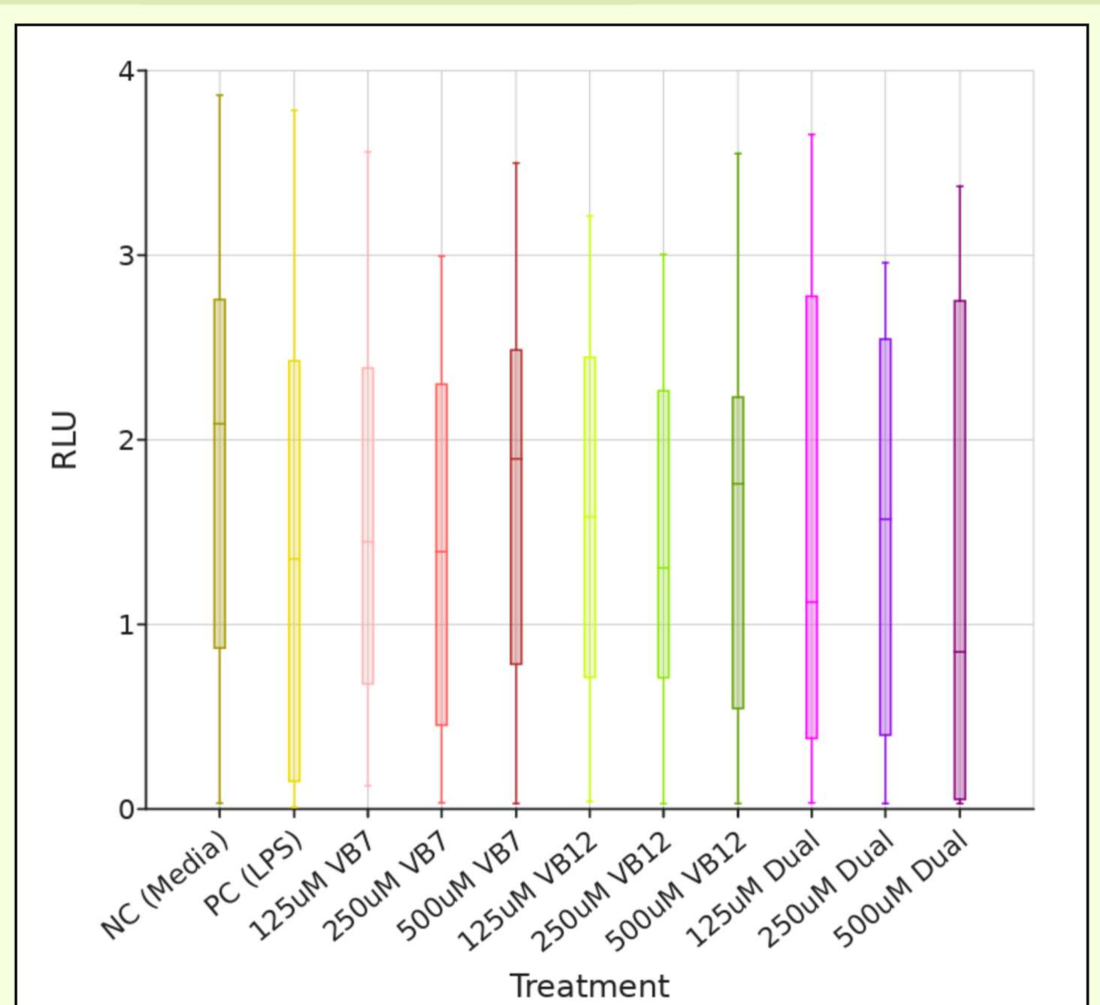


Figure 3. Relative Light Units vs. Treatment Groups

Degrees of Freedom (df)	H-statistic	P-value	Interpretation of P
10	8.48	0.58	A P-value of **0.58** means **no evidence** that the groups might be different.

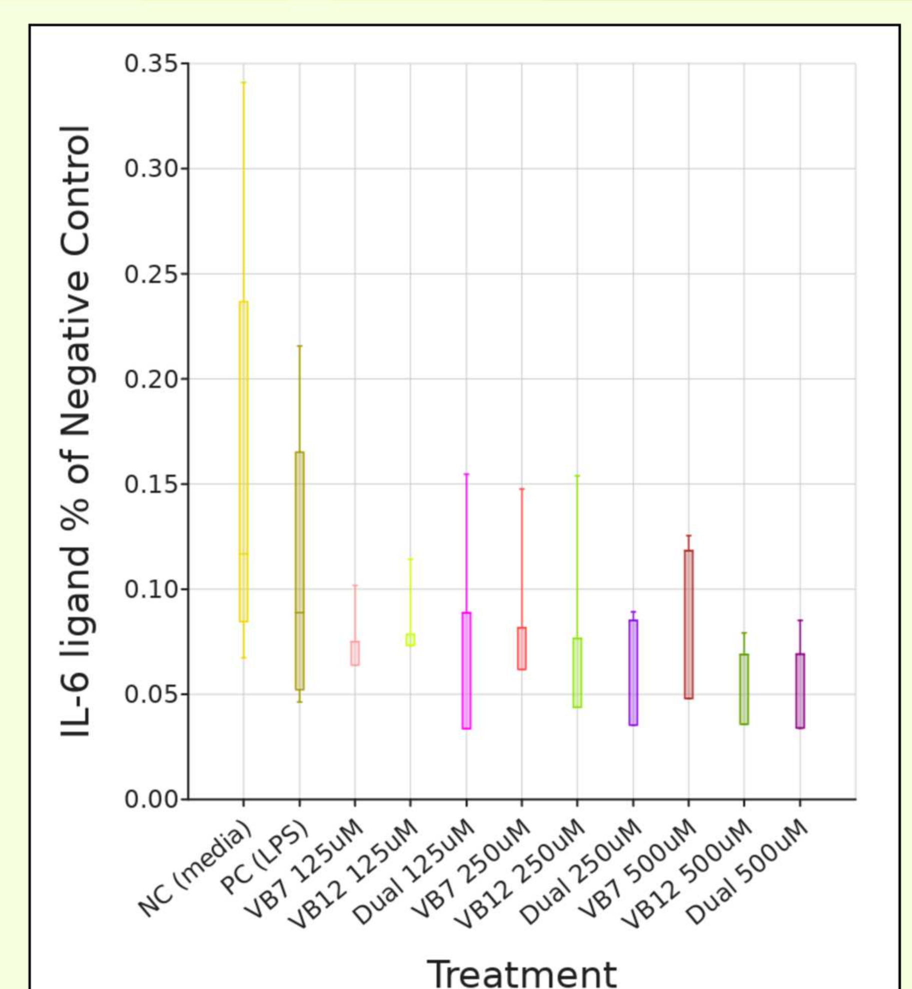


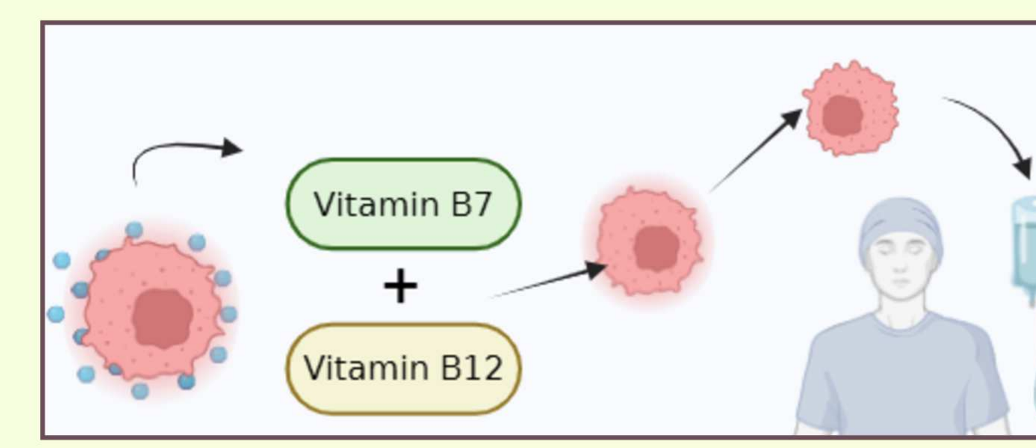
Figure 4. IL-6 Presence as % of Control vs. Treatment Groups

Degrees of Freedom (df)	H-statistic	P-value	Interpretation of P
10	9.28	0.51	A P-value of **0.51** means **no evidence** that the groups might be different.

Discussion

- Relative light units of treatments were excepted to have little to no statisitcal difference if they were not toxic to the cells, given the results the **treatments are not toxic to the cells**
- A slight decrease in relative light units was expected from negative to positive control; however this was not found to be true, an indicator the **LPS concentration was not strong enough**
- **IL-6 Presence** did not increase following LPS Stimulation, indicating the concentration must be increased
- Due to the LPS concentration being insufficient to stimulate the cells, **no conclusion** can be made on the effect of treatments on LPS stimulated cells.
  - However, IL-6 Presence was not statistically different across any treatment, meaning they **did not stimulate IL-6** when LPS is present at a low dosage.

Next Steps



- Increase LPS Concentration, with potential concentration curve
- Test treatments with new LPS Concentration

References

Greten, F. R., & Grivennikov, S. I. (2019). Inflammation and Cancer: Triggers, Mechanisms, and Consequences. Immunity, 51(1), 27-41. https://doi.org/10.1016/j.immuni.2019.06.025

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Wang, C., Xiu, Y., Zhang, Y., Wang, Y., Xu, J., Yu, W., & Xing, D. (2025). Recent advances in biotin-based therapeutic agents for cancer therapy. Nanoscale, 1(4). https://doi.org/10.1039/d4nr03729d

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